

Informe de puerta E8 — run_v1.22_r3 (Paper IV v1.22-r3)

Veredicto actual: **PASS_WITH_OBSERVATIONS**. Evidencia: 20_EVIDENCE/E8/. Auditor: Claude Opus 5.5, misma sesión que las ejecuciones anteriores; no se ejecutó Lean.

E8 — Supplementary verifier, X-28, E8-02 and attribution (r3)

Verifier inspection (`verify_frozen_logs.py`, sha256 in the r3 manifest)

- **Fail-closed design.** It requires the inventory. A directory whose `.log` set differs from the expected one (missing or extra files) fails. A hash mismatch fails. An alert (sorry or error) fails. A total count $\neq$ `expected_total` fails. It writes its output only to a new file (`open('x')`) and does not run Lean. It checks logs, not mathematical validity or the kernel.
- **Sorry rule.** `\bsorryAx\b|declaration\s+uses\s+["'"]*sorry\bwithre.Γ`. It is stronger than the frozen rules:
 - `audit_publication.py:52` uses `sorryAx|declaration` uses 'sorry' and is case-sensitive;
 - `paperiv_build.py:400` uses the literal "declaration uses 'sorry'".
- **Error rule (E8-02).** Pattern `^. *error(?:\(|:)` with `re.M` and **without** `re.I`. It is literally the error alternative of the frozen `audit_publication.py`. On the probe table (`E8_R3_CHECK.json`), "ERROR:" and "Error:" are not alerted by either rule, so **case sensitivity is identical** to the historical rule. The sorry reinforcement uses a separate regex and does not change the error rule. **E8-02 CLOSED** (the concern about adding `re.I` to errors does not apply).

Repetitions with the auditor's own output

1. **In place (read-only).** `python verify_frozen_logs.py --inventory LOG_INVENTORY.json --output <run_v1.22_r3>/20_EVIDENCE/E8/rerun_in_place/VERIFY_FROZEN_LOGS_AUDITOR_OUTPUT.json`: **PASS, 1266 logs checked, 0 errors**, exit 0. Running it wrote nothing in the r3 folder; the pre-existing `__pycache__` keeps its 10:56 timestamp.
2. **Test copy** (`20_EVIDENCE/E8/testcopy/`). The verifier and harness were copied. The only change is the Paper IV root path in the copy of the verifier, documented in `testcopy_verifier_patch.diff`; the harness is unchanged. `test_log_verifier.py` reproduced:
 - **15/15 unit cases:** the real negative log; backtick, quote and curly-quote variants; uppercase; line breaks; `[sorryAx]`; 3 negatives (`no_sorry_used`, `sorryish`, clean axioms); 2 lowercase errors; and uppercase `ERROR`: not alerted.
 - **4/4 CLI corruptions** (sorry, error, changed hash, missing file): exit 1 and FAIL in every case.
 - The historical scan: PASS, 1266.
 - Comparison with the sealed files: the regenerated inventory equals `LOG_INVENTORY.json`; the unit cases equal `LOG_VERIFIER_TESTS.json`; the CLI outcomes and the historical scan are identical. The sealed package was not overwritten.
3. **Independent recount** (`10_SCRIPTS/e8_r3_check.py`).
 - Four groups: $608 + 50 + 422 + 186 = 1266$, with 0 duplicates and 0 hash mismatches.
 - These are the same four directories as the E8 scan of `run_v1.21_r1` and `run_v1.22_r2`.
 - The same 1266 files have now been scanned four times (`v1.21`, `r2`, verifier in place, test copy). **They are counted once:** 1266 files, not 1266 proofs.

Historical nuance

The frozen `audit_publication.py` rule does detect the real negative log, but only through `sorryAx` in an axiom `printout('AuditorNeg.uses_sorry' depends on axioms: [sorryAx])`. It does not detect the warning itself, which Lean 4.28 prints with backticks. `paperiv_build.py` does not detect it at all.

X-28 decision

The supplementary verifier is designated as a **mandatory control** of the r3 reproduction profile (`CORRECTIONS_AND_HANDOFF_v1.22_r3.md`). It is fail-closed, pinned by hash and reproducible, and its tests pass.

- **The control action of the delivery is CLOSED.**
- **The historical observation is kept:** the frozen runners still carry their insufficient pattern; they were not patched and this is not claimed.
- The axiom and cone evidence (18 headline declarations, 50 public cones) remains the substantive check against `sorryAx`.
- X-28 → MITIGATED, with the control in place. **No action is needed before publication**, provided the reproduction profile cites the verifier as mandatory.

Other items

- 1266/844 attribution (E8-01): the auditor's scan covers 1266; the author's old script covered $844 = 1266 - 422$. Resolved in r1; kept.
- Editor documents (`ARTIFACT_CHECKS`, `VISUAL_REVIEW`, `STATUS_DELTA`, `EDITORIAL_INTEGRITY_REPORT`, `CORRECTIONS_AND_HANDOFF`) were contrasted with the auditor's own evidence: the page lists, diff, protected content, 615 entries and figures all agree. They are not used as an inherited PASS.
- The claim «Los párrafos de estado conservan paridad» is correct in meaning; the ES terminology issue is NEW-05 (E6).

E8 verdict (r3): PASS_WITH_OBSERVATIONS. X-28's historical observation is kept; the control is closed.

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